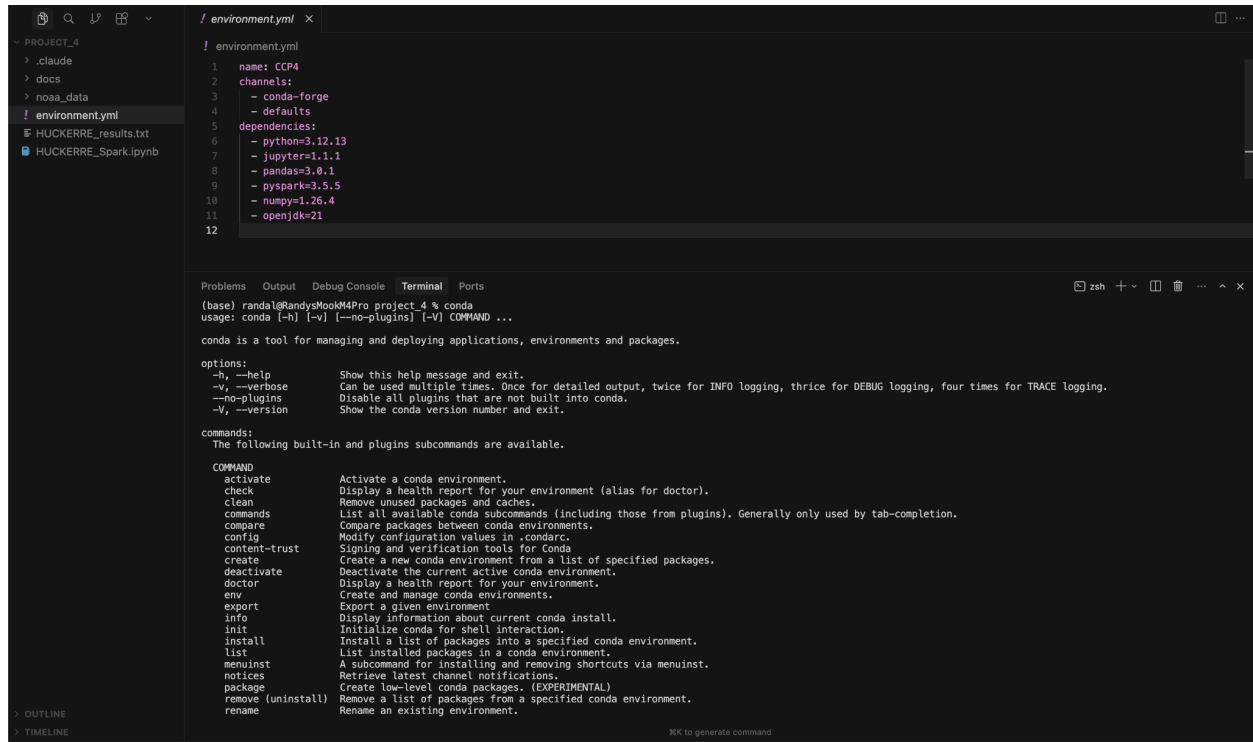


Questions:

1. Download Anaconda Python, install PySpark, and launch Jupyter Notebook (Points: 1):

- Provide screenshots of each major step (Anaconda Python setup, PySpark pip install, and launching Jupyter Notebook or Jupyter Lab).



The screenshot shows an IDE with a file explorer on the left containing 'PROJECT_4', '.claude', 'docs', 'noaa_data', 'environment.yml', 'HUCKERRE_results.txt', and 'HUCKERRE_Spark.ipynb'. The main editor displays the 'environment.yml' file with the following content:

```
! environment.yml
1 name: CCP4
2 channels:
3   - conda-forge
4   - defaults
5 dependencies:
6   - python=3.12.13
7   - jupyter=1.1.1
8   - pandas=3.0.1
9   - pyspark=3.5.5
10  - numpy=1.26.4
11  - openjdk=21
12
```

Below the editor is a terminal window showing the output of the 'conda' command:

```
(base) randal@RandysMookM4Pro project_4 % conda
usage: conda [-h] [-v] [--no-plugins] [-V] COMMAND ...

conda is a tool for managing and deploying applications, environments and packages.

options:
  -h, --help            Show this help message and exit.
  -v, --verbose          Can be used multiple times. Once for detailed output, twice for INFO logging, thrice for DEBUG logging, four times for TRACE logging.
  --no-plugins          Disable all plugins that are not built into conda.
  -V, --version          Show the conda version number and exit.

commands:
  The following built-in and plugins subcommands are available.

COMMAND
activate      Activate a conda environment.
check         Display a health report for your environment (alias for doctor).
clean         Remove unused packages and caches.
commands     List all available conda subcommands (including those from plugins). Generally only used by tab-completion.
compare       Compare packages between conda environments.
config        Modify configuration values in .condarc.
content-trust Signing and verification tools for Conda
create        Create a new conda environment from a list of specified packages.
deactivate    Deactivate the current active conda environment.
doctor        Display a health report for your environment.
env           Create and manage conda environments.
export        Export a given environment
info          Display information about current conda install.
init          Initialize conda for shell interaction.
install       Install a list of packages into a specified conda environment.
list          List installed packages in a conda environment.
menuinst      A subcommand for installing and removing shortcuts via menuinst.
notices       Retrieve latest channel notifications.
package       Create low-level conda packages. (EXPERIMENTAL)
remove (uninstall) Remove a list of packages from a specified conda environment.
rename        Rename an existing environment.
```

```
● (base) randal@RandysMookM4Pro project_4 % conda env list

# conda environments:
#
# * -> active
# + -> frozen
base          *  /Users/randal/miniconda3
CCP4          *  /Users/randal/miniconda3/envs/CCP4
```

Project 4 - NOAA Weather Data Analysis with PySpark

Analyzing NCEI Global Surface Summary of Day data for Cincinnati (72429793812) and Florida (99495199999) from 2015-2024.

Setup - Initialize Spark Session

```
from pyspark.sql import SparkSession
from pyspark.sql import functions as F
from pyspark.sql.window import Window
from pyspark.sql.types import DoubleType, IntegerType, StringType
import os

# Initialize Spark session
spark = SparkSession.builder \
    .appName("Project4_WeatherAnalysis") \
    .master("local[*]") \
    .getOrCreate()

spark
```

Question 2 - Load CSV files and display the count of each dataset

There should be 19 results (no Florida data for 2016).

```
# Define base path and station info
```

Problems Output Debug Console Terminal Ports

```
#
# * -> active
# + -> frozen
base          * /Users/ranald/miniconda3
CCP4          /Users/ranald/miniconda3/envs/CCP4
```

OUTLINE

(base) randal@RandysMookM4Pro project_4 %

Setup - Initialize Spark Session

```
from pyspark.sql import SparkSession
from pyspark.sql import functions as F
from pyspark.sql.window import Window
from pyspark.sql.types import DoubleType, IntegerType, StringType
import os

# Initialize Spark session
spark = SparkSession.builder \
    .appName("Project4_WeatherAnalysis") \
    .master("local[*]") \
    .getOrCreate()

spark
```

Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/03/24 22:21:56 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

SparkSession - in-memory

SparkContext

Spark UI

Version v3.5.5

Master local[*]

AppName Project4_WeatherAnalysis

2. Load the CSV files and display the count of each dataset (Points: 1):

- Submit the code and output. There should be 19 results. (weather data for Florida (station #99495199999) does not exist for the year 2016)

```
2015 - Cincinnati (72429793812): 365 records
2015 - Florida (99495199999): 355 records
2016 - Cincinnati (72429793812): 366 records
2016 - Florida (99495199999): FILE DOES NOT EXIST
2017 - Cincinnati (72429793812): 365 records
2017 - Florida (99495199999): 283 records
2018 - Cincinnati (72429793812): 365 records
2018 - Florida (99495199999): 363 records
2019 - Cincinnati (72429793812): 365 records
2019 - Florida (99495199999): 345 records
2020 - Cincinnati (72429793812): 366 records
2020 - Florida (99495199999): 365 records
2021 - Cincinnati (72429793812): 365 records
2021 - Florida (99495199999): 104 records
2022 - Cincinnati (72429793812): 365 records
2022 - Florida (99495199999): 259 records
2023 - Cincinnati (72429793812): 365 records
2023 - Florida (99495199999): 276 records
2024 - Cincinnati (72429793812): 366 records
2024 - Florida (99495199999): 133 records
```

Total datasets loaded: 19

3. Find the hottest day (column MAX) for each year (Points: 1):
- Provide the corresponding station code, station name, date, and temperature (columns: STATION, NAME, DATE, MAX).
 - There should be 10 results.

STATION	NAME	DATE	MAX
72429793812	CINCINNATI MUNICIPAL AIRPORT LUNKEN FIELD, OH US	2015-06-12	91.9
72429793812	CINCINNATI MUNICIPAL AIRPORT LUNKEN FIELD, OH US	2016-07-24	93.9
72429793812	CINCINNATI MUNICIPAL AIRPORT LUNKEN FIELD, OH US	2017-07-22	91.9
72429793812	CINCINNATI MUNICIPAL AIRPORT LUNKEN FIELD, OH US	2018-07-04	96.1
72429793812	CINCINNATI MUNICIPAL AIRPORT LUNKEN FIELD, OH US	2019-09-30	95.0
72429793812	CINCINNATI MUNICIPAL AIRPORT LUNKEN FIELD, OH US	2020-07-05	93.9
72429793812	CINCINNATI MUNICIPAL AIRPORT LUNKEN FIELD, OH US	2021-08-12	95.0
72429793812	CINCINNATI MUNICIPAL AIRPORT LUNKEN FIELD, OH US	2022-06-14	96.1
72429793812	CINCINNATI MUNICIPAL AIRPORT LUNKEN FIELD, OH US	2023-08-23	96.1
72429793812	CINCINNATI MUNICIPAL AIRPORT LUNKEN FIELD, OH US	2024-08-30	100.9

4. Find the coldest day (column MIN) for the month of March across all years (2015-2024) (Points: 1):

- Provide the corresponding station code, station name, date, and temperature (columns: STATION, NAME, DATE, MIN).
- There should be 1 result.

STATION	NAME	DATE	MIN
72429793812	CINCINNATI MUNICIPAL AIRPORT LUNKEN FIELD, OH US	2015-03-06	3.2

5. Find the year with the most precipitation for Cincinnati and Florida (Points: 1):

- Provide the corresponding station code, station name, and year (columns: STATION, NAME, YEAR, Mean of PRCP).
- There should be 2 results.

STATION	NAME	YEAR	Mean of PRCP
72429793812	CINCINNATI MUNICIPAL AIRPORT LUNKEN FIELD, OH US	2018	0.15789041095890405
99495199999	SEBASTIAN INLET STATE PARK, FL US	2015	0.0

Note - there was no precipitation data from Florida.

6. Count the percentage of missing values for wind gust (column GUST) for Cincinnati and Florida in the year 2024 (Points: 1):

- There should be 2 results.

STATION	NAME	total_records	missing_count	missing_percentage
72429793812	CINCINNATI MUNICIPAL AIRPORT LUNKEN FIELD, OH US	366	143	39.07
99495199999	SEBASTIAN INLET STATE PARK, FL US	133	133	100.0

7. Find the mean, median, mode, and standard deviation of the temperature (column TEMP) for Cincinnati in each month for the year 2020 (Points: 1):

- There should be 12 results (one for each month, with 4 values for each result).

MONTH	Mean	Median	Mode	Std_Dev
1	37.95	37.7	24.7	8.35
2	36.59	36.0	25.9	7.9
3	49.07	47.8	39.6	8.78
4	51.78	51.0	39.2	7.31
5	60.89	63.7	73.9	9.31
6	72.55	73.7	70.7	4.9
7	77.6	77.9	72.5	2.34
8	73.35	73.7	67.4	3.49
9	66.1	65.8	54.7	7.12
10	55.19	54.0	41.4	6.73
11	48.0	47.7	47.7	6.83
12	35.99	35.2	32.1	6.64

Note - My mode conflicts take the lower value.

8. Find the top 10 days with the lowest Wind Chill for Cincinnati in 2017 (Points: 1):

- For days where TEMP is below 50°F and WDSP (wind speed) is above 3 mph, calculate Wind Chill using the formula:

$$WC = 35.74 + 0.6215 \times TEMP - 35.75 \times (WDSP)^{0.16} + 0.4275 \times TEMP \times (WDSP)^{0.16}$$

- Add a new column for Wind Chill and display the top 10 days with the lowest Wind Chill.

STATION	NAME	DATE	TEMP	WDSP	Wind_Chill
72429793812	CINCINNATI MUNICIPAL AIRPORT LUNKEN FIELD, OH US	2017-01-07	10.5	7.0	-1.38
72429793812	CINCINNATI MUNICIPAL AIRPORT LUNKEN FIELD, OH US	2017-12-31	11.0	5.3	1.11
72429793812	CINCINNATI MUNICIPAL AIRPORT LUNKEN FIELD, OH US	2017-12-27	13.0	5.8	2.91
72429793812	CINCINNATI MUNICIPAL AIRPORT LUNKEN FIELD, OH US	2017-12-28	13.6	5.8	3.63
72429793812	CINCINNATI MUNICIPAL AIRPORT LUNKEN FIELD, OH US	2017-01-06	13.6	5.5	3.98
72429793812	CINCINNATI MUNICIPAL AIRPORT LUNKEN FIELD, OH US	2017-01-08	15.9	5.2	7.07
72429793812	CINCINNATI MUNICIPAL AIRPORT LUNKEN FIELD, OH US	2017-12-25	25.8	13.5	13.43
72429793812	CINCINNATI MUNICIPAL AIRPORT LUNKEN FIELD, OH US	2017-12-30	21.6	5.3	13.75
72429793812	CINCINNATI MUNICIPAL AIRPORT LUNKEN FIELD, OH US	2017-01-05	22.2	5.8	13.96
72429793812	CINCINNATI MUNICIPAL AIRPORT LUNKEN FIELD, OH US	2017-12-26	23.3	6.2	14.9

Note - I converted the WDSP from Knots to Mph as instructed. The conversion is Mph = Knots * 1.15078

9. Investigate how many days had extreme weather conditions for Florida (fog, rain, snow, etc.) using the FRSHTT column (Points: 1).

- There should be 1 result.
- Details can be found in the [README.pdf](#)

Total days with extreme weather conditions for Florida: 0						
Fog	Rain	Snow	Hail	Thunder	Tornado	
0	0	0	0	0	0	

10. Predict the maximum Temperature for Cincinnati for November and December 2024, based on the previous 2 years of weather data (Points: 1):

- There should be 2 results.
- You can use any model to forecast on the historical weather data.

- Submit the code, prediction results, and a brief discussion on the model's performance and potential improvements.

```
26/03/24 22:33:42 WARN InstanceBuilder: Failed to load implementation from:dev.ludovic.netlib.blas.JNIBLAS
26/03/24 22:33:42 WARN InstanceBuilder: Failed to load implementation from:dev.ludovic.netlib.blas.VectorBLAS
Model Coefficients: [0.07845602401354168,0.0,1.4586710562621248,-0.49853899792871387]
Model Intercept: 9.529992234630255
Training RMSE: 4.27
Training R2: 0.9372
```

Predicted vs Actual MAX Temperature for Cincinnati Nov & Dec 2024:

MONTH	Predicted_MAX	Actual_MAX
11	85.43	80.1
12	75.06	63.0

Test RMSE: 5.01

Test R2: 0.8141